

BANARAS HINDU UNIVERSITY

B.Voc. Semester III Examination, 2023-24

Paper: BVCA-302 — Robotics with Arduino

Time: Three Hours

Maximum Marks: 70

Instructions: Section A: Attempt any 2 out of 4 questions (12 marks each). Section B: Attempt any 6 out of 9 questions (6 marks each). Section C: All 10 questions compulsory (1 mark each).

SECTION — A

Note: Long-answer questions to be answered in about 500 words each. Attempt any two. $[12 \times 2 = 24]$

Question 1. Explain the fundamental components and functions of both hydraulic and pneumatic systems used in robotics. Provide a detailed comparison between the two systems, highlighting their similarities and differences. Additionally, discuss the advantages and limitations of each system in the context of robotics applications.

Question 2. Discuss the components and operation of a machine vision system used in robotics. Explain the role of lighting, cameras, frame grabbers, and computer processors in capturing and processing visual data for robot guidance and control. Provide examples of machine vision applications in industrial automation and robotics.

Question 3. Explain the basic components of a hydraulic system and their functions. Provide examples of applications where hydraulic systems are commonly used.

Question 4. Describe the operation of a pneumatic system, including its basic components and their functions. Discuss how a compressor generates compressed air, and how pneumatic actuators convert this energy into mechanical power.

SECTION — B

Note: Attempt any six of the following questions. Answer in about 200 words each. $[6 \times 6 = 36]$

Question 5. Define the term “robotics” and explain its significance in modern industrial automation. Provide examples of industries where robotics is commonly utilized, highlighting its impact on productivity and quality improvement.

Question 6. Explain the concept of degrees of freedom in industrial robots. Describe how degrees of freedom are related to the robot’s ability to perform various tasks and navigate its work envelope efficiently.

Question 7. Explain the historical development of robotics from its origins in science fiction to its modern applications in industries. Highlight key milestones and technological advancements that have shaped the field.

Question 8. Discuss the advantages and disadvantages of different drive systems used in industrial robots, including hydraulic drive, electric drive, and pneumatic drive. Provide examples of scenarios where each drive system is preferred based on performance requirements.

Question 9. Define actuators and sensors in the context of robotics. Explain their respective roles in robotic systems and provide examples of each. Discuss how actuators and sensors work together to enable robotic motion and perception in industrial and non-industrial applications.

Question 10. Discuss the importance of precision in robot movement and its components: spatial resolution, accuracy, and repeatability.

Question 11. Differentiate between fixed automation, programmable automation, and flexible automation. Provide examples of industries where each type is commonly employed, and discuss their respective advantages and limitations.

Question 12. Explain the concept of ROS (Robot Operating System) and its significance in robotics development.

Question 13. Describe the various types of motors used in robotics, including DC motors, AC motors, stepper motors, and servomotors.

SECTION — C

Note: Answer all objective-type questions.

[10 × 1 = 10]

Question 14. Answer the following:

- (i) In robot motion planning, what is the purpose of the “configuration space”?
 - (a) The physical space occupied by the robot
 - (b) The space of all possible joint configurations of the robot
 - (c) The workspace where the robot performs tasks
 - (d) The space-time continuum in which the robot operates
- (ii) What is the role of the “end effector” in a robot manipulator?
 - (a) To control the robot’s motion
 - (b) To provide feedback to the operator
 - (c) To act as the hand or tool of the robot
 - (d) To process sensor data
- (iii) Which type of robot programming involves specifying the desired end-effector positions and orientations without detailing the intermediate joint motions?
 - (a) Motion Planning
 - (b) Trajectory Planning
 - (c) Inverse Kinematics

- (d) Forward Kinematics
- (iv) What does the term “ROS package” refer to in the Robot Operating System (ROS)?
 - (a) A physical package containing robot components
 - (b) A collection of software libraries, tools, and configuration files
 - (c) The payload carried by a robot
 - (d) A unit of measurement for robot speed
- (v) Which concept in robotics involves a robot’s ability to recover from errors or unexpected events without human intervention?
 - (a) Fault Tolerance
 - (b) Redundancy
 - (c) Closed-Loop Control
 - (d) Safety Interlock
- (vi) In the context of swarm robotics, what does the term “stigmergy” refer to?
 - (a) Communication through direct signalling
 - (b) Cooperation based on the use of pheromones or environmental cues
 - (c) Hierarchical control structure
 - (d) Swarm navigation using GPS
- (vii) What is the origin of the term “robot”?
 - (a) Greek
 - (b) Czechoslovakia
 - (c) German
 - (d) Latin
- (viii) When did modern industrial robotics technology begin?
 - (a) 1920
 - (b) 1960
 - (c) 1980
 - (d) 2000
- (ix) How is robotics defined with respect to industrial automation?
 - (a) A subset of automation
 - (b) Unrelated to automation
 - (c) Only involves computer programming

- (d) A replacement for automation
- (x) What is another name for the Cartesian coordinate configurable robot?
 - (a) Spherical robot
 - (b) XYZ robot
 - (c) Rectangular robot
 - (d) Hyperbolic robot